Lab Exercise 16- Creating a ConfigMap from a Folder

In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

Step 1: Set Up Kubernetes Cluster

Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.

Step 2: Prepare the Folder with Multiple Files

Create a directory named configmap-folder:

```
mkdir configmap-folder
cd configmap-folder/
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> mkdir configmap-folder


    Directory: C:\Users\Devanshi\Desktop\COS_Lab\lab16


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        02-03-2026     22:23                configmap-folder


PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> cd configmap-folder/
```

Create multiple configuration files inside the directory:

config1.txt with content:

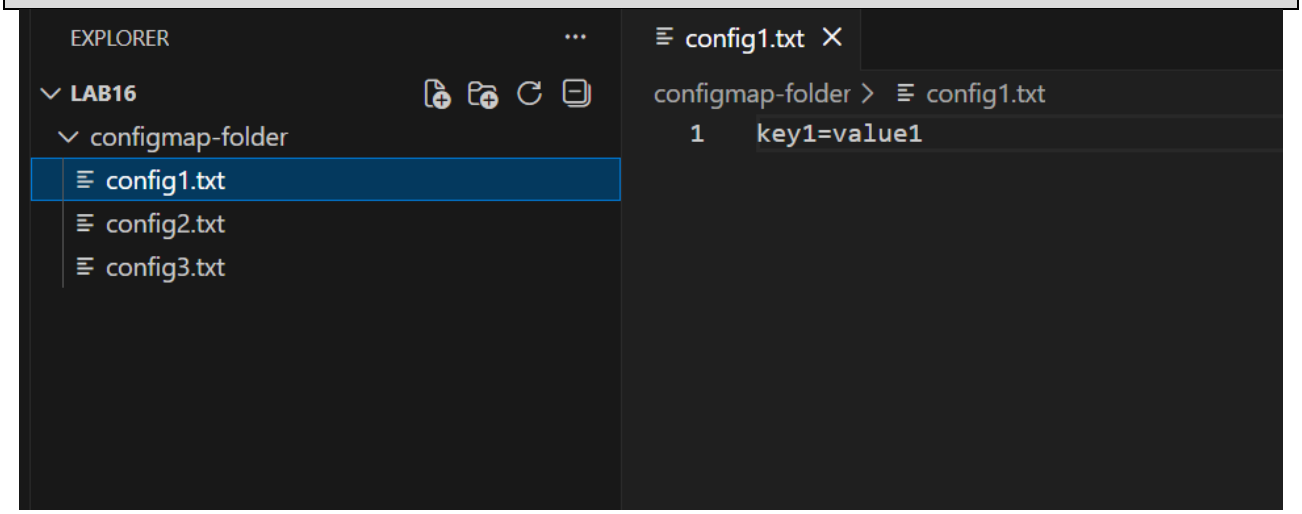
```
key1=value1
```

config2.txt with content:

```
key2=value2
```

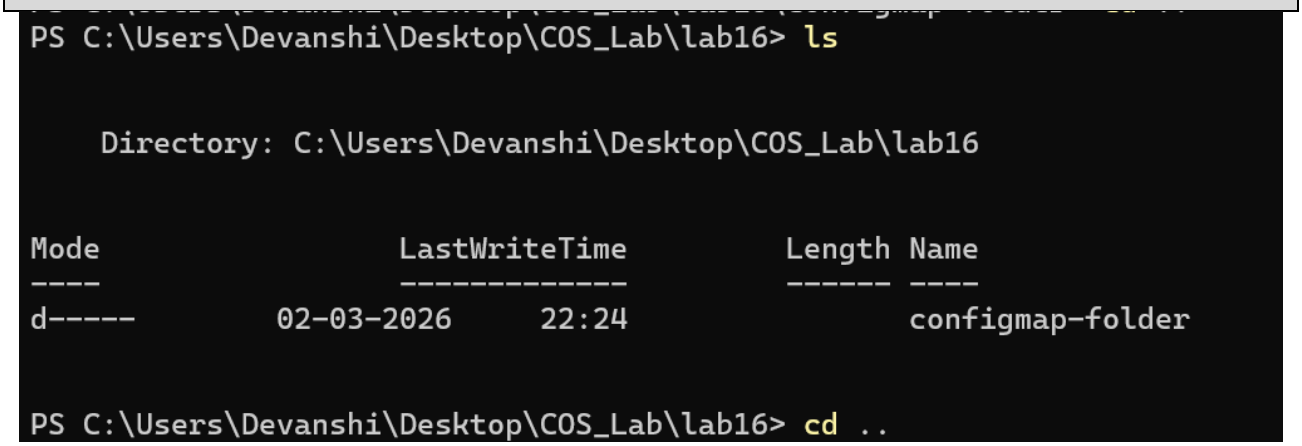
config3.txt with content:

```
key3=value3
```



Verify the directory structure:

```
ls configmap-folder/
cd ..
```



You should see the files config1.txt, config2.txt, and config3.txt.

Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```

Verify the ConfigMap:

```
kubectl get configmap folder-configmap -o yaml
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl create configmap folder-configmap --from-file=configmap-folder/
configmap/folder-configmap created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl get configmap folder-configmap -o yaml
apiVersion: v1
data:
  config1.txt: key1=value1
  config2.txt: key2=value2
  config3.txt: key3=value3
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-02T16:58:11Z"
  name: folder-configmap
  namespace: default
  resourceVersion: "2493"
  uid: f38f360b-099d-487e-af7a-b8ff89ba6ae6
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
        name: folder-configmap
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-folder-configmap.yaml
```

Check the status of the Pod:

```
kubectl get pods
```

View the Pod logs to see the content of the mounted files:

```
kubectl exec -it configmap-pod -- printenv
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl apply -f pod-config.yaml
pod/configmap-pod created
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl get pods
NAME            READY   STATUS    RESTARTS   AGE
configmap-pod   1/1     Running   0          63s
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16>
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl exec -it configmap-pod -- printenv
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=configmap-pod
TERM=xterm
config3.txt=key3=value3
config1.txt=key1=value1
config2.txt=key2=value2
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
NGINX_VERSION=1.29.5
NJS_VERSION=0.9.5
NJS_RELEASE=1~trixie
ACME_VERSION=0.3.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
HOME=/root
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

Step 5: Clean Up

After completing the exercise, clean up the resources created:

```
kubectl delete pod folder-configmap-pod
kubectl delete configmap folder-configmap
rm -r configmap-folder
```

```
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl delete pod folder-configmap-pod
Error from server (NotFound): pods "folder-configmap-pod" not found
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl delete configmap folder-configmap
configmap "folder-configmap" deleted
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> rm -r configmap-folder
PS C:\Users\Devanshi\Desktop\COS_Lab\lab16> kubectl delete pod configmap-pod
pod "configmap-pod" deleted
```